

MICHIGAN CITY MUNICIPAL, IN, USA

WMO: 726358

Lat: 41.703N

Lon: 86.821W

Elev: 200

StdP: 98.94

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 00384

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.6	-15.1	-22.9	0.5	-16.8	-20.1	0.6	-13.3	11.0	-3.4	9.8	-5.1	4.0	230	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	31.4	22.7	30.1	21.9	27.9	20.9	24.4	29.0	23.5	27.7	22.7	26.6	3.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.1	25.5	22.4	17.5	25.2	22.0	17.0	25.0	74.8	29.1	71.3	28.2	68.1	27.3	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.2	8.0	7.1		DB	-21.0	33.1	5.9	2.2	-25.2	34.7	-28.7	36.0	-32.0	37.2	-36.3	38.7
				WB	-21.4	25.9	5.9	1.2	-25.6	26.8	-29.0	27.5	-32.3	28.2	-36.6	29.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-4.1	-2.1	2.9	8.5	14.7	19.2	21.5	20.6	18.2	11.6	4.3	0.4	(6)
	DBStd	10.21	6.53	6.30	6.03	5.04	5.52	4.20	3.47	2.86	4.41	4.86	5.03	5.34	(7)
	HDD10.0	1643	437	341	237	87	17	0	0	0	1	40	182	301	(8)
	HDD18.3	3506	695	573	482	299	143	41	11	11	58	217	421	556	(9)
	CDD10.0	1531	0	1	15	41	162	276	357	330	246	88	12	3	(10)
	CDD18.3	351	0	0	2	2	30	66	109	82	53	7	0	0	(11)
	CDH23.3	2833	0	0	12	23	277	588	921	597	373	41	1	0	(12)
	CDH26.7	778	0	0	1	3	65	169	292	137	106	6	0	0	(13)
(14) Wind	WSAvg	2.9	3.8	3.5	3.3	3.3	2.7	2.3	2.0	1.6	1.8	3.0	3.6	3.3	(14)
(15) Precipitation	PrecAvg	998	65	53	64	90	94	110	103	101	86	89	78	65	(15)
	PrecMax	1442	146	107	131	172	174	254	184	248	346	226	190	126	(16)
	PrecMin	759	20	20	21	27	16	38	40	22	18	21	14	21	(17)
	PrecStd	156	32	25	29	39	47	54	37	58	66	52	37	26	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	17.0	26.1	26.2	31.1	32.3	35.9	32.2	32.2	27.4	21.1	16.2	(19)
		MCWB	12.0	12.4	17.1	16.0	19.9	21.7	23.0	23.0	22.6	21.0	15.0	14.4	(20)
	2%	DB	9.8	12.6	20.0	22.8	27.9	30.3	32.0	30.1	30.1	23.9	17.3	13.7	(21)
		MCWB	8.7	10.1	15.0	14.3	18.7	20.7	23.7	22.3	23.3	17.9	13.4	11.9	(22)
	5%	DB	6.1	9.1	16.1	20.9	27.0	28.1	29.9	27.7	27.4	21.7	14.9	10.1	(23)
		MCWB	4.8	7.2	12.0	14.0	18.2	20.0	22.5	21.4	21.6	17.5	10.3	8.0	(24)
	10%	DB	3.5	6.1	12.4	17.6	24.4	27.2	27.6	27.0	25.2	19.8	12.1	7.4	(25)
		MCWB	2.3	3.7	8.8	11.4	17.2	19.4	21.2	21.0	20.3	16.5	9.2	5.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	14.0	17.7	17.4	21.8	24.0	26.1	25.2	25.7	22.0	16.3	15.4	(27)
		MCDB	13.5	16.0	24.9	23.2	28.0	30.3	30.7	28.5	28.8	24.9	18.7	16.1	(28)
	2%	WB	9.1	10.9	15.8	15.8	20.5	22.9	24.5	23.8	23.8	19.8	14.0	12.0	(29)
		MCDB	9.8	12.8	20.4	20.1	25.8	27.5	29.8	27.5	27.8	22.9	17.5	13.2	(30)
	5%	WB	5.3	7.2	12.8	14.3	19.5	22.1	23.6	22.8	22.7	18.2	11.2	8.0	(31)
		MCDB	6.7	9.1	14.7	18.9	24.6	26.7	28.1	26.3	26.3	20.4	13.4	8.9	(32)
	10%	WB	2.9	3.8	9.3	12.2	18.5	20.9	22.8	22.1	21.5	17.0	9.0	5.9	(33)
		MCDB	3.6	5.9	12.3	17.4	23.3	25.1	26.7	25.4	24.2	18.9	11.8	7.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.4	8.4	10.3	11.0	11.1	10.2	10.5	10.6	9.1	7.2	6.1	(35)
		MCDBR	8.5	11.5	13.1	14.8	13.4	12.6	12.1	12.0	11.7	11.4	10.7	9.3	(36)
	5% WB	MCWBR	7.8	9.7	8.6	8.8	6.6	6.0	5.2	5.9	5.7	7.3	7.1	7.7	(37)
		MCWBR	8.8	11.4	11.8	13.1	11.5	11.0	10.9	9.6	10.1	9.9	10.0	8.3	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.306	0.328	0.346	0.400	0.430	0.435	0.449	0.446	0.420	0.358	0.323	0.304	(40)
		taud	2.426	2.348	2.374	2.251	2.228	2.263	2.221	2.227	2.261	2.438	2.488	2.485	(41)
	Ebn at Noon	Ebn at Noon	843	874	900	869	849	843	826	816	810	831	818	817	(42)
		Edh at Noon	78	99	108	132	138	134	139	134	121	90	74	69	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.50	2.34	3.56	4.65	5.30	6.08	6.13	5.32	4.38	2.85	1.77	1.26	(44)
	RadStd	RadStd	0.15	0.30	0.28	0.39	0.47	0.38	0.30	0.31	0.36	0.30	0.23	0.14	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page